Conditional Statements — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

☞ *Video: If p, then q*

Key Idea

If it rains, then the ground is wet.

p = the **hypothesis** (the "if" part) q = the **conclusion** (the "then" part)

Written: $p \to q$

☞ *Video: Four related statements*

Key Idea

Conditional	$p \to q$	If it rains, then the ground is wet.
Converse	$q \to p$	**switch**: If the ground is wet, then it rained.
Inverse	$\sim p \to \sim q$	**negate**: If it does not rain, then the ground is not wet.
Contrapositive	$\sim q \to \sim p$	**switch and negate**: If the ground is not wet, then it did not rain.

☞ *Video: True or false?*

Watch Me — Worked Example

Conditional: If it rains, the ground is wet. **true**

Converse: If the ground is wet, it rained. **false** — counterexample: a **sprinkler**

Contrapositive: If the ground is not wet, it did not rain. **true**

☞ *Video: The rule*

Key Idea

A conditional and its **contrapositive** are always both true or both false.

The **converse** and the **inverse** also match each other — but they may not match the original.

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☞ *Video: Biconditional*

Key Idea

If a conditional **and** its converse are both true, write "__if and only if__".

Example: An angle is right if and only if it measures __$90°$__.

☞ *Video: Your turn*

Your Turn

"If an angle measures $30°$, then it is acute." Write the **converse**. Is it true?

Converse: If an angle is __acute__, then it measures __$30°$__.

True or false? __false__ Counterexample: a __$50°$__ angle is acute. (any acute angle except $30°$)

Notice

Remember: Converse **switches**, inverse **negates**, contrapositive does **both**. A conditional and its contrapositive are both true or both false. Both directions true? Use "if and only if."

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